BILLBOARD HIT CLASSIFICATION

Can audio features identify Billboard hits?

I tested whether 12 musical descriptors could distinguish songs with a recorded Billboard Hot 100 appearance in a historical benchmark. I trained on older releases and reserved the 2000s for the final test.

0.678
Test ROC-AUC

70.3%
Hit recall

44.3%
Hit precision

What happened on 2,413 test songs

Metric	Random Forest	Always non-hit
Accuracy	61.3%	67.2%
Balanced accuracy	63.6%	50.0%
Hit recall	70.3%	0.0%
Hit precision	44.3%	Undefined
ROC-AUC	0.678	0.500

WHAT A REVIEW TEAM WOULD SEE

The model found 557 of 792 labelled hits and flagged 699 non-hits. That is a modest historical ranking signal with a substantial review burden.

MODEL SELECTION & TEST ERRORS

How I tested the model

The split follows release year. Chart position defines the label. Release year, artist identity, chart fields and listening counts were excluded from the predictors.

Partition	Release years	Songs	Hit share
Training	1960–1989	2,907	70.11%
Validation	1990–1999	1,961	37.02%
Test	2000–2009	2,413	32.82%

The Random Forest led validation ROC-AUC at 0.670, ahead of Logistic Regression at 0.633. I chose the 0.65 threshold on validation balanced accuracy and kept it fixed on the test set.

Test confusion matrix

	Predicted non-hit	Predicted hit
Actual non-hit	922	699
Actual hit	235	557

Balanced accuracy averages recall for hits and non-hits. The threshold is a decision rule for this benchmark. It has no calibrated probability interpretation.

INTERPRETATION & LIMITATIONS

What the signal can support

Which features the forest relied on

The largest validation ROC-AUC drops after shuffling individual inputs came from the happy mood score (0.0273), dynamic complexity (0.0215), and dissonance (0.0168). This measures the fitted model's reliance on those features. It gives no direction or causal effect.

How the benchmark limits interpretation

HSP-S samples hits and non-hits, so the class shares do not estimate industry hit rates. The 287 songs without matched release years were disproportionately hits. Also, 236 test songs included a training-period artist.

Why this is a historical classification test

The split uses release years and the chart labels are retrospective. A track's eventual chart appearance might not have been known at the time represented by its split. The results do not establish performance for today's releases.

What I would test next

Use a fixed post-release outcome window and features available before release. Compare rolling and artist-disjoint splits, evaluate calibration, and choose a decision threshold using review capacity and the cost of false alarms.

DATA SOURCE

HSP-S · Vötter, Mayerl, Specht & Zangerle · Zenodo